

CHAPTER 1

PLANTS AROUND US

You have learned in class one that plants grow all around us. Can you name some of the plants that grow around you, at home, at school, in your neighborhood park or on your farm?

Plants are one of the many beautiful gifts of nature. They are very useful for us. We eat some plants and we use some plants to make other things.

ACTIVITIES

1. Write the names of three (3) plants around you.

(i) _____ (ii) _____ (iii) _____

2. Write the names of the plants in given spaces.



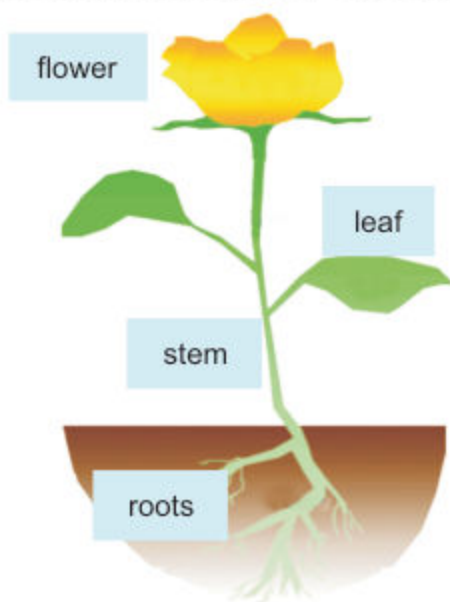
CHAPTER 2

PARTS OF PLANTS AND THEIR FUNCTIONS

You have learnt in class one that plants have many parts.

Can you name the parts of a plant?

Yes, plants have flowers, leaves, stems and roots.



Each part of the plant helps the plant in some way. In the next few pages you will learn how each part of the plant helps it grow.

ACTIVITY

Draw a plant. Label the parts of a plant and colour it.



The Root

All plants have roots. The roots grow down into the soil.

They have many small branches.

The functions of the root are as follows:

1. Roots help to keep the plant fixed in the same place.



ACTIVITY

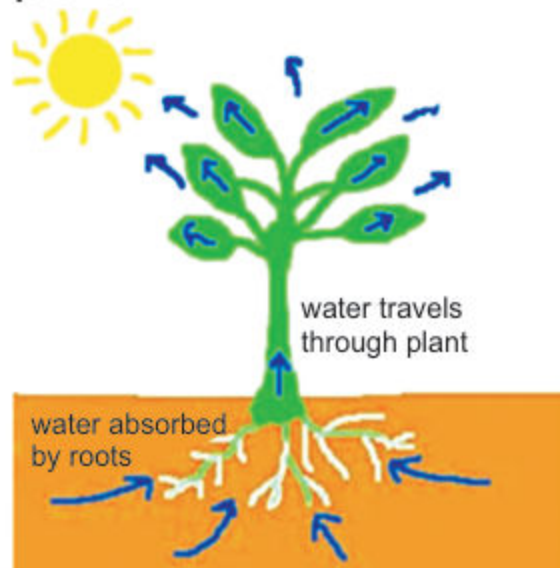
Bring in a plant with its roots intact. Observe the roots. Shake off most of the dirt. Observe the roots again. Answer the questions.

- (I) What did you observe? _____
- (II) Why is there dirt on the roots? _____
- (III) Why do roots grow underground? _____
- (IV) Why do roots have small branches? _____

Teacher's Note Explain to students that roots are underground. They spread out so as to hold the plant firmly in its place. Explain how roots work sucking up water and nutrients from the ground and giving it to the other parts of the plant.

Expected answer for activity; The water will move up the paper towel.

2. Roots take water and nutrients from the soil and give it to the whole plant.



ACTIVITY

Things Needed

- Strip of cotton cloth or paper towel
- A cup of water

Procedure:

Fill a cup with water. Dip one end of the strip of cotton cloth or paper towel in the water. Observe what happens.

What did you observe happen? _____

3. Some roots store food in them. We eat some of the roots that store food in them such as carrot, raddish and *arvi*.



Raddish



Carrot



Arvi

ACTIVITIES

1. Fill in the blanks .Use the word box to help you.

food grow roots support

- (i) Roots _____ into the soil.
 - (ii) Water and nutrients are absorbed by the ____ of a plant.
 - (iii) We eat some roots as _____.
 - (iv) Roots give _____ to the plant to stand straight.
2. Draw three roots that you eat, name and colour them.

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The Stem

The stem is the part of the plant that is between the roots and the leaves.

The functions of the stem are as follows:



- 1. The stem keeps the plant standing straight.
- 2. The stem supports branches, leaves, flowers and fruits.
- 3. It takes water from the roots and gives it to the leaves to make food.
- 4. Some plants store food in their stems. We eat the stems of many plants such as sugarcane.

Teacher's Note Explain to the students that it is the stem's job to carry water to the leaves and flowers of plants. It works like a straw, sucking the water up.

ACTIVITY

Things needed:

- A glass
- lettuce leaves/cabbage with stem
- Food colour (red or blue, ink can also be used)
- Water



Procedure:

Half fill a glass with water. Place a lettuce leaf (with stem) in a glass of water. Add the food colour to the water and mark the water level on the glass. Over the next three days observe what happens to the lettuce leaf and the water. Write your observations in the table below.

Day	Level of water in the glass	Colour of the lettuce leaf
1.		
2.		
3.		

Answer the following questions.

(i) What was the colour of the lettuce leaf in the beginning?

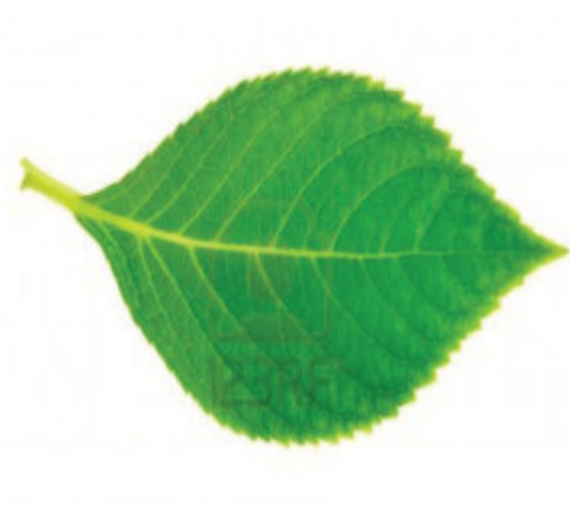
(ii) What was the colour of the lettuce leaf at the end of the experiment?

(iii) Why did the colour of the lettuce leaf change?

(iv) What is the function of the stem?

The Leaf

Leaves grow on the stem and its branches. Some leaves are simple, that is, a single leaf is attached to the stem. Some leaves are compound, that is, many leaves are attached to a stem.



Simple leaf



Compound leaf

Leaves are mostly green in colour. Some leaves stay green all year round. They are evergreen. Others turn brown and then fall off the stem.

Leaves are of many different shapes and sizes. Some leaves are needle like. These leaves are evergreen. Some leaves are plain and some leaves have margins (teeth).



Needle like leaf



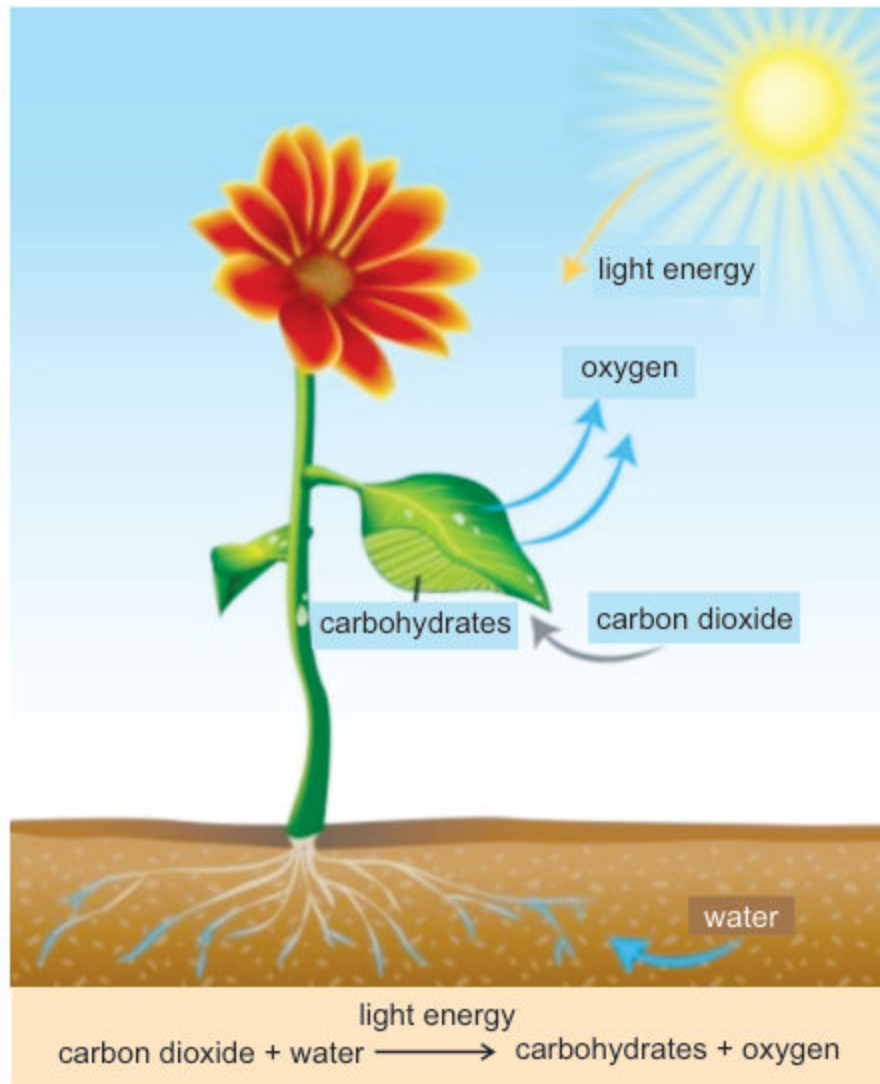
Plain leaf



Margins (Teeth leaf)

The functions of the leaves are as follows:

1. The leaves make the food for the plant. They use water, carbon dioxide from the air and sunlight to make the food.



2. We eat the leaves of some plants such as spinach, mint, coriander and *methi*.



Spinach



Mint



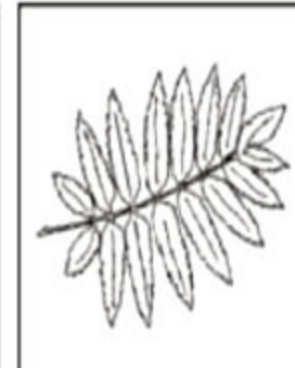
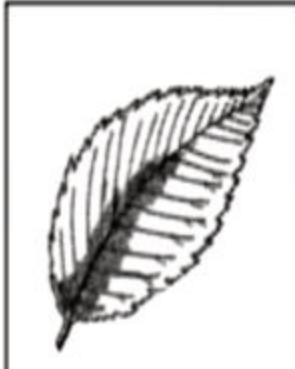
Coriander



Methi

ACTIVITY

In the table below, different types of leaves are given. Put a tick (✓) on number 1 for simple leaves, on number 2 for compound leaves number 3 for needle leaves.

			
1 2 3	1 2 3	1 2 3	1 2 3
			
1 2 3	1 2 3	1 2 3	1 2 3
			
1 2 3	1 2 3	1 2 3	1 2 3

Flowers

Flowers are the colourful and beautiful part of the plants.

The functions of flowers are as follows.

1. The colourful and pleasant smell of flowers attracts butterflies and bees to plants. The butterflies and bees help in pollination.



A butterfly and a honeybee on flowers

2. Flowers also make the many fruits that we eat such as mangoes and oranges. Seeds are present in the fruit. New plants grow from the seeds.



Mango tree with flower

Flowers with fruit

Mango fruit

Mango seed

Teacher's Note Explain why the flowers are brightly coloured. Show the students where to find the pollen and let them observe it with a magnifying glass. Explain how they play a role in pollination. Tell them about pollination and the flower's job of making seeds.

3. Flowers make our environment beautiful.



Flowering and non-flowering plants

Not all the plants have flowers. The plants that produce flowers and seeds are called flowering plants.



The plants that do not produce flowers and seeds are called non-flowering plants.



ACTIVITIES

1. Answer the following questions.

(i) Why are the flowers brightly colored?

(ii) Did you see any insect, butterflies or bees on the flowers?
What were the insects doing?

(iii) What are the main functions of flowers?

2. Draw how a flower becomes a fruit with seeds and seeds grow into a new plant in the given box.



Guava flower



Guava fruit



Guava seed



A new guava plant

END OF UNIT 8: EXERCISES

1a. Draw a plant. Colour and label its parts.



b. Write the things that a plant needs to grow.

(i) _____

(ii) _____

(iii) _____

(iv) _____

2. Fill in the blanks .Use the word box to help you.

flower ,fruit , trunk, water, food

(i) Roots take in _____ from the soil.

(ii) The stem of a tree is called the_____.

(iii) The colorful part of a plant is the_____.

(iv) Leaves make _____ for the plants.

(v) Flowers make _____.

2. Look at the pictures and answer the questions.



(i) Write the names of the plants.

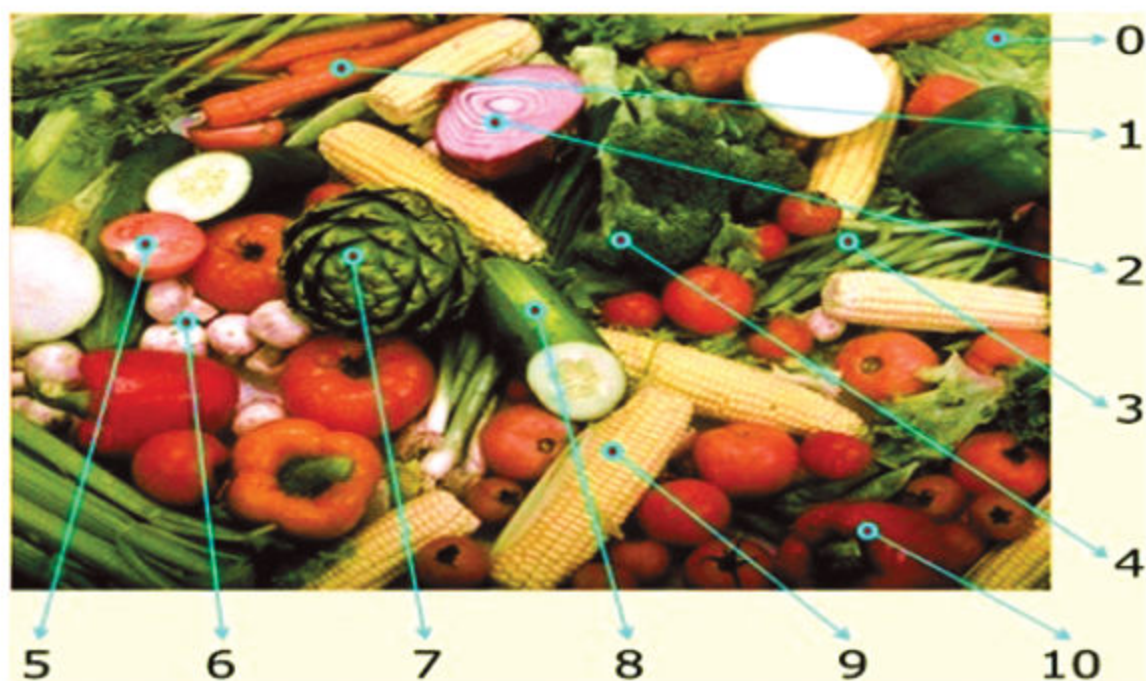
1. _____

2. _____

(ii) In what ways are these plants similar?

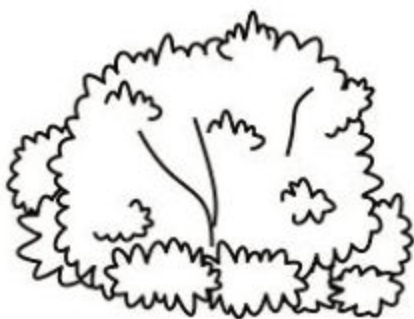
iii) What are the differences between these two plants?
(Find as many differences as you can. Look at the leaves,
stem, branches, flowers and fruits (if any).)

- 3.a Look at the picture below. Some vegetables are given numbers from 0 to 10. Identify which part of the plant each vegetable is. Complete the table below.



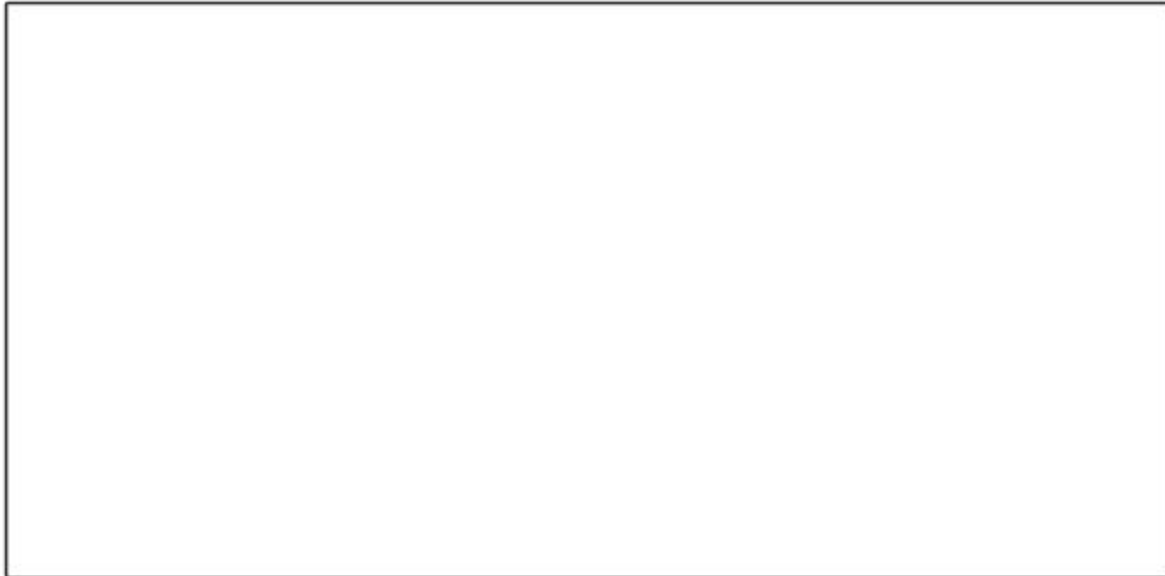
No	Name of the vegetable	Part of plant
0		
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

- b. Pictures of some plants are given. Write in the given space flowering or non flowering.



EXTENSION ACTIVITIES

1. Draw your favorite fruit with seed.



2. List the ways plants are useful for us? Use the word box help you.

Clean air, food, shelter for birds, shade for us, wood, beautiful environment, furniture, paper, cloth, medicine, fuel to burn, cotton.

Plants give	We make from plants

Teacher's Note Explain more clearly the importance of plants and difference between what we get from plants and what we make from plants.

3. Write the names of two fruit you eat that have seeds and of two fruit which have no seeds.

a. Two fruits with seeds:

(i) _____ (ii) _____

b. Two fruits without seeds:

(i) _____ (ii) _____

4. Take two leafy plants. Carefully remove the leaves from one plant. Place both plants in a sunny spot and water and care for them the same way over the next few weeks. If the plant without leaves grows new ones, remove them too. Observe what happens to the two plants and write it down.

Yes, without the leaves, the plant can't make food and dies.